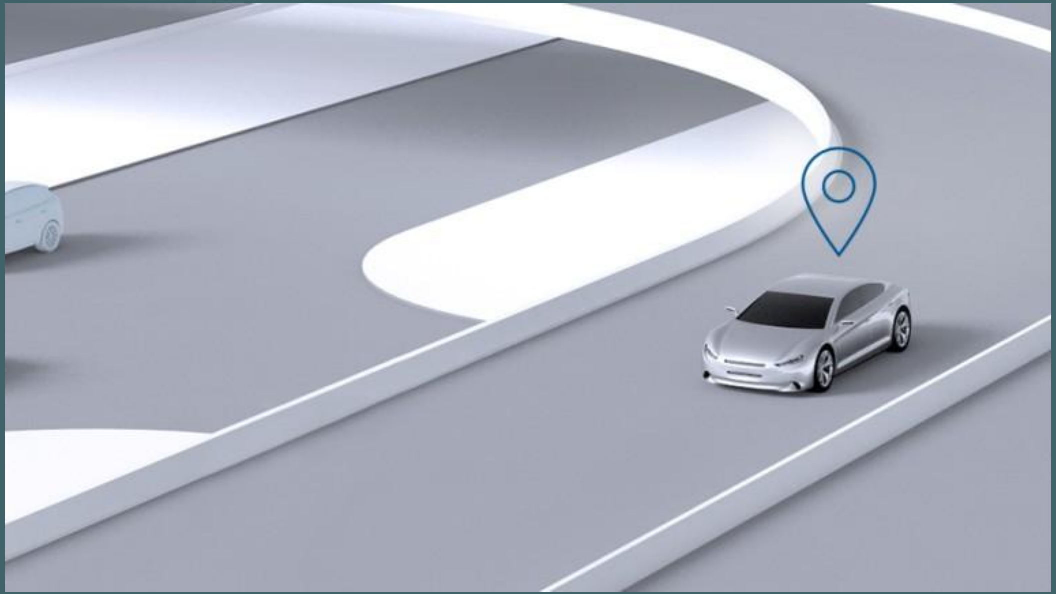


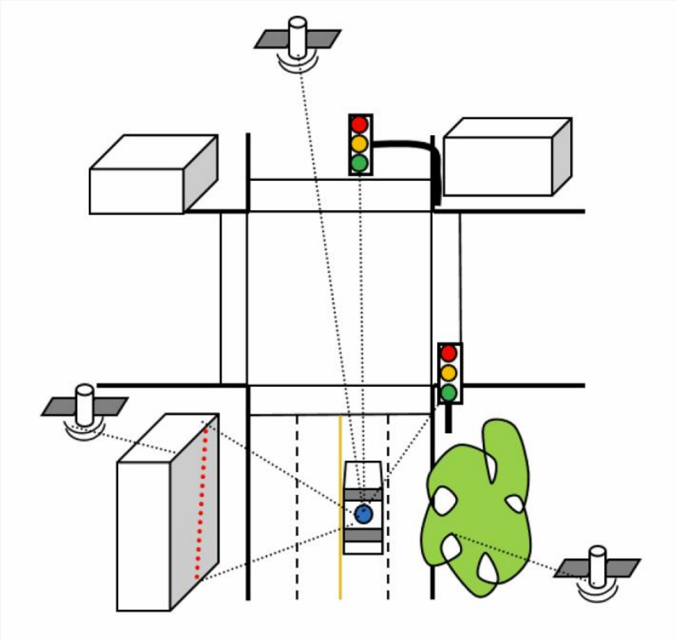
Adaptive Sensor Fusion for Vehicle Localization

Presented by- Anirudh Gudi
SID- 862548787

Introduction

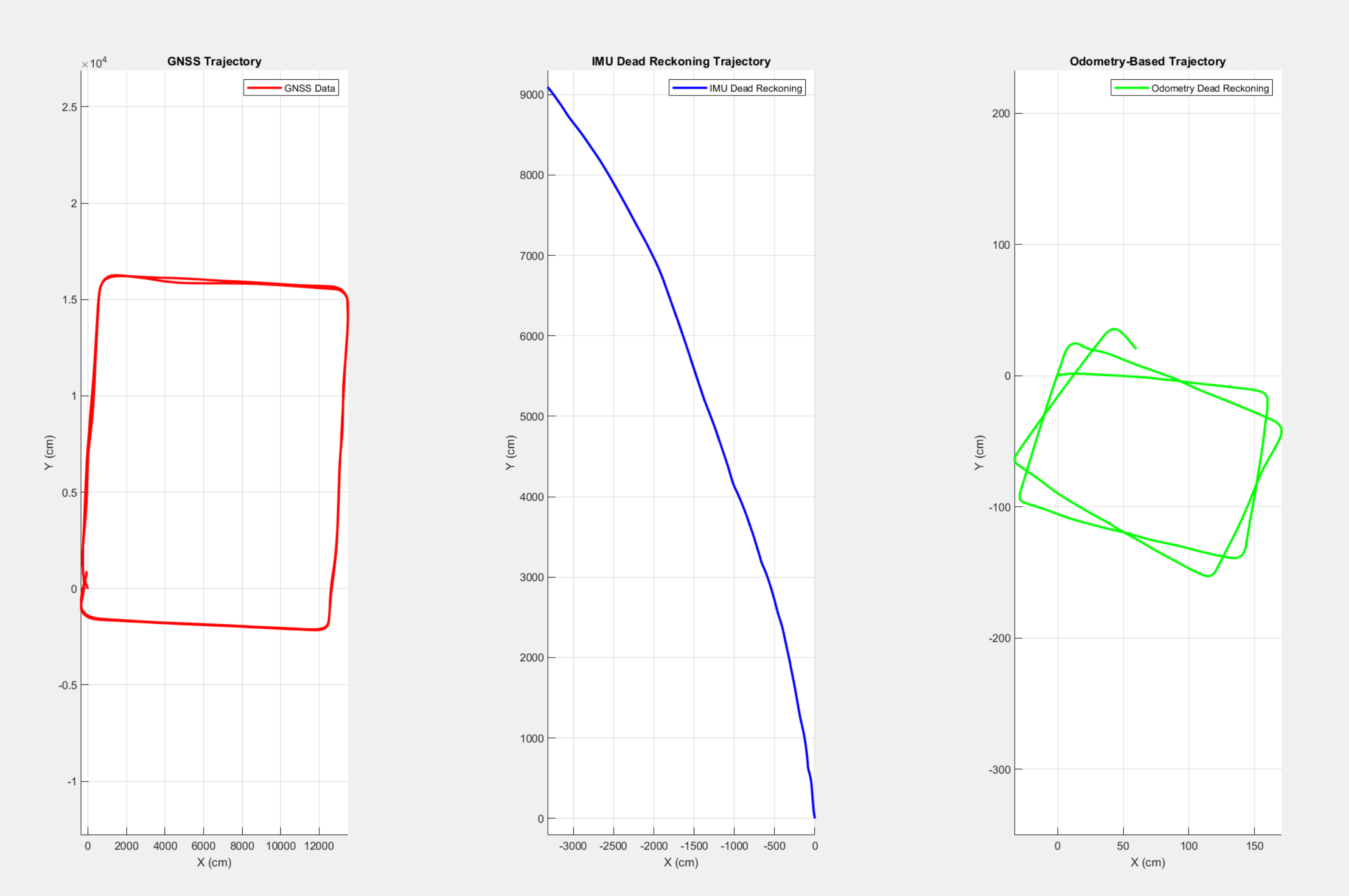
- Vehicle localization is crucial for autonomous navigation.
- Sensors such as GNSS, IMU, and Odometry have limitations.
- Sensor fusion enhances localization accuracy.
- Proposing an adaptive sensor fusion approach.

Motivation



- GNSS is unreliable in urban areas.
- IMU suffers from drift over time.
- Odometry is affected by wheel slip.
- Adaptive fusion selects the best filter dynamically.

Motivation

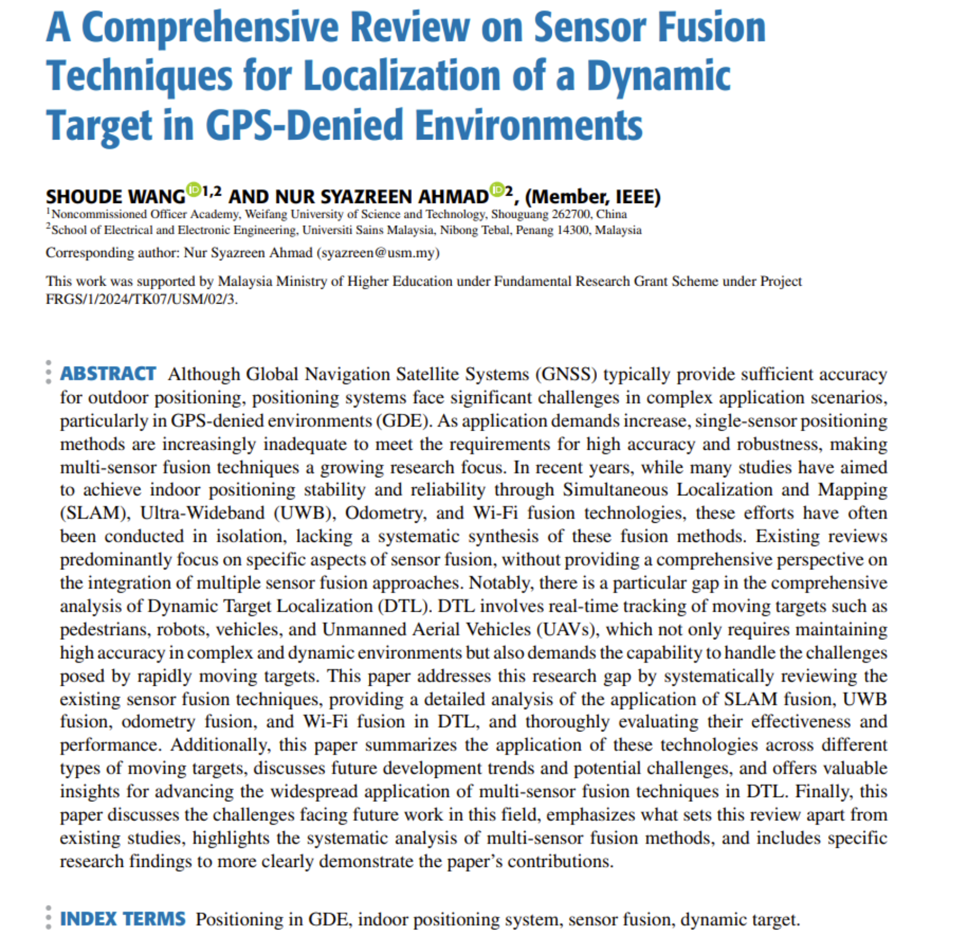


GNSS Trajectory
GNSS Data
×10^4
Y (cm)
X (cm)
IMU Dead Reckoning Trajectory
IMU Dead Reckoning
Y (cm)
X (cm)
Odometry-Based Trajectory
Odometry Dead Reckoning
Y (cm)
X (cm)

Literature Review

- GNSS-based localization: Prone to signal errors.
- IMU-based dead reckoning: Prone to drift.
- Kalman Filters (EKF, UKF) for fusion.
- Particle Filters (PF) for non-Gaussian distributions.
- Factor Graph Optimization (FGO) for batch optimization.

Literature Review



A Comprehensive Review on Sensor Fusion Techniques for Localization of a Dynamic Target in GPS-Denied Environments

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ABSTRACT Although Global Navigation Satellite Systems (GNSS) typically provide sufficient accuracy for outdoor positioning, positioning systems face significant challenges in complex application scenarios, particularly in GPS-denied environments (GDE). As application demands increase, single-sensor positioning methods are increasingly inadequate to meet the requirements for high accuracy and robustness, making multi-sensor fusion techniques a growing research focus. In recent years, while many studies have aimed to achieve indoor positioning stability and reliability through Simultaneous Localization and Mapping (SLAM), Ultra-Wideband (UWB), Odometry, and Wi-Fi fusion technologies, these efforts have often been conducted in isolation, lacking a systematic synthesis of these fusion methods. Existing reviews predominantly focus on specific aspects of sensor fusion, without providing a comprehensive perspective on the integration of multiple sensor fusion approaches. Notably, there is a particular gap in the comprehensive analysis of Dynamic Target Localization (DTL). DTL involves real-time tracking of moving targets such as pedestrians, robots, vehicles, and Unmanned Aerial Vehicles (UAVs), which not only requires maintaining high accuracy in complex and dynamic environments but also demands the capability to handle the challenges posed by rapidly moving targets. This paper addresses this research gap by systematically reviewing the existing sensor fusion techniques, providing a detailed analysis of the application of SLAM fusion, UWB fusion, odometry fusion, and Wi-Fi fusion in DTL, and thoroughly evaluating their effectiveness and performance. Additionally, this paper summarizes the application of these technologies across different types of moving targets, discusses future development trends and potential challenges, and offers valuable insights for advancing the widespread application of multi-sensor fusion techniques in DTL. Finally, this paper discusses the challenges facing future work in this field, emphasizes what sets this review apart from existing studies, highlights the systematic analysis of multi-sensor fusion methods, and includes specific research findings to more clearly demonstrate the paper's contributions.

INDEX TERMS Positioning in GDE, indoor positioning system, sensor fusion, dynamic target.

Literature Review

Factor graph optimization for GNSS/INS integration: A comparison with the extended Kalman filter

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Abstract
Factor graph optimization (FGO) recently has attracted attention as an alternative to the extended Kalman filter (EKF) for GNSS-INS integration. This study evaluates both loosely and tightly coupled integrations of GNSS code pseudorange and INS measurements for real-time positioning, using both conventional EKF and FGO with a dataset collected in an urban canyon in Hong Kong. The FGO strength is analyzed by degenerating the FGO-based estimator into an “EKF-like estimator.” In addition, the effects of window size on FGO performance are evaluated by considering both the GNSS pseudorange error models and environmental conditions. We conclude that the conventional FGO outperforms the EKF because of the following two factors: (1) FGO uses multiple iterations during the estimation to achieve a robust estimation; and (2) FGO better explores the time correlation between the measurements and states, based on a batch of historical data, when the measurements do not follow the Gaussian noise assumption.

KEYWORDS
extended Kalman filter, factor graph optimization, GNSS, INS, integration, navigation, positioning, urban canyons, window size

Problem Formulation

- Goal: Develop an adaptive sensor fusion technique.
- Challenges:
  - Quantifying sensor confidence scores.
  - Dynamic switching between filters.
  - Handling high-noise conditions.
- Proposed Solution: Confidence-driven filter selection.

Proposed Approach

Proposed Approach

- Define confidence scores (GNSS, IMU, Odometry, Sensor Availability).
- Compute weighted confidence score.
- Select the best filter dynamically:
  - EKF for high confidence.
  - UKF for moderate confidence.
  - PF for low confidence.
  - FGO for extreme noise.

Proposed Approach

Define confidence scores for each sensor:

- **CS_GNSS**: Measures GNSS reliability based on noise levels.
- **CS_IMU**: Captures IMU drift.
- **CS_Odom**: Reflects odometry reliability based on speed variation.
- **CS_Sensor**: Checks sensor availability (valid data).

```
% Using a window (last 10 samples or fewer)
window = max(i-10, 1):i;

% Compute GNSS Confidence Score (Low variance → High Confidence)
gnss_var = var(z_noisy(1:2, max(i-10,1):i), 0, 2);
max_gnss = max(gnss_var);
if max_gnss == 0, max_gnss = 1e-6; end  % Prevent division by zero
CS_GNSS = exp(-gnss_var / max_gnss);

% Compute IMU Confidence Score (Higher variance → Lower Confidence)
imu_vals = [accel_x_all(max(i-10,1):i), accel_y_all(max(i-10,1):i), gyro_z_all(max(i-10,1):i)];
imu_drift = var(imu_vals(:));
if imu_drift == 0, imu_drift = 1e-6; end  % Prevent division by zero
CS_IMU = exp(-imu_drift / max(imu_drift));

% Compute Odometry Confidence Score (More variation in speed → Lower Confidence)
odom_var = var(speed_interp(max(i-10,1):i));
CS_Odom = exp(-odom_var / max(odom_var));

% Compute Sensor Availability Confidence
CS_Sensor = double(~any(isnan(z)));
```

Proposed Approach

```
% Final Confidence Score Calculation
CS_final = 0.5 * CS_GNSS + 0.2 * CS_IMU + 0.25 * CS_Sensor + 0.1 * CS_Odom;
```

```
% Select filter based on confidence score.
if CS_final > 0.7
    selected_filter = 'EKF';  % High Confidence → EKF
elseif CS_final > 0.5
    selected_filter = 'UKF';  % Moderate Confidence → UKF
elseif CS_final > 0.3
    selected_filter = 'PF';   % Low Confidence → PF
else
    selected_filter = 'FGO';  % Very Low Confidence → FGO
```

Implementation and Results

Implementation

- Dataset: GNSS, IMU, Odometry data.
- Preprocessing: Synchronization and noise injection.
- Adaptive Sensor Fusion Algorithm:

  - State Initialization.

  - Confidence Score Computation.

  - Filter Selection.

  - State Update.
- Implemented in MATLAB.

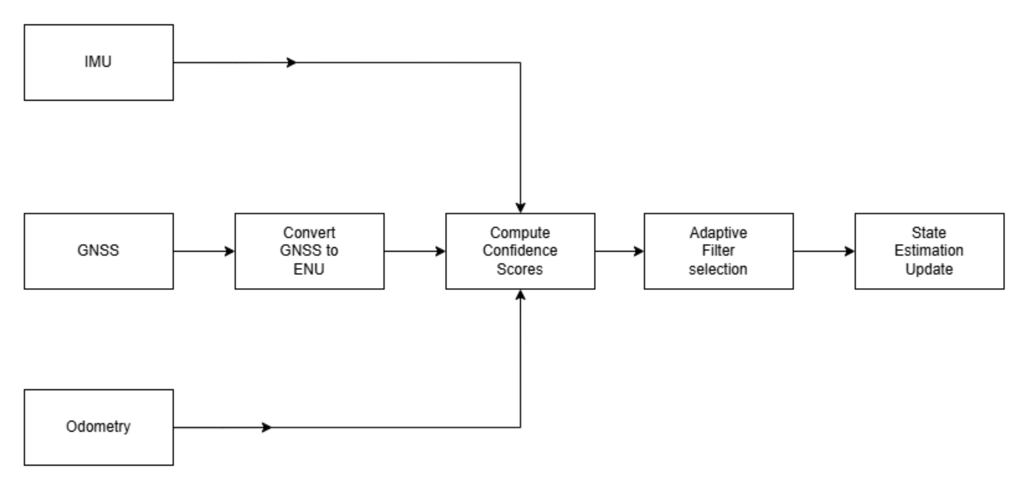
1.3. Dataset 2: UrbanNav-HK-Data20200314

Brief: Dataset UrbanNav-HK-Data2020314 is collected in a low-urbanization area in Kowloon which suitable for algorithmic verification and comparison. The coordinates transformation between multiple sensors, and intrinsic measurements of camera can be found via Extrinsic Parameters, IMU Nosie and Intrinsic Parameters of Camera.

Some key features are as follows:

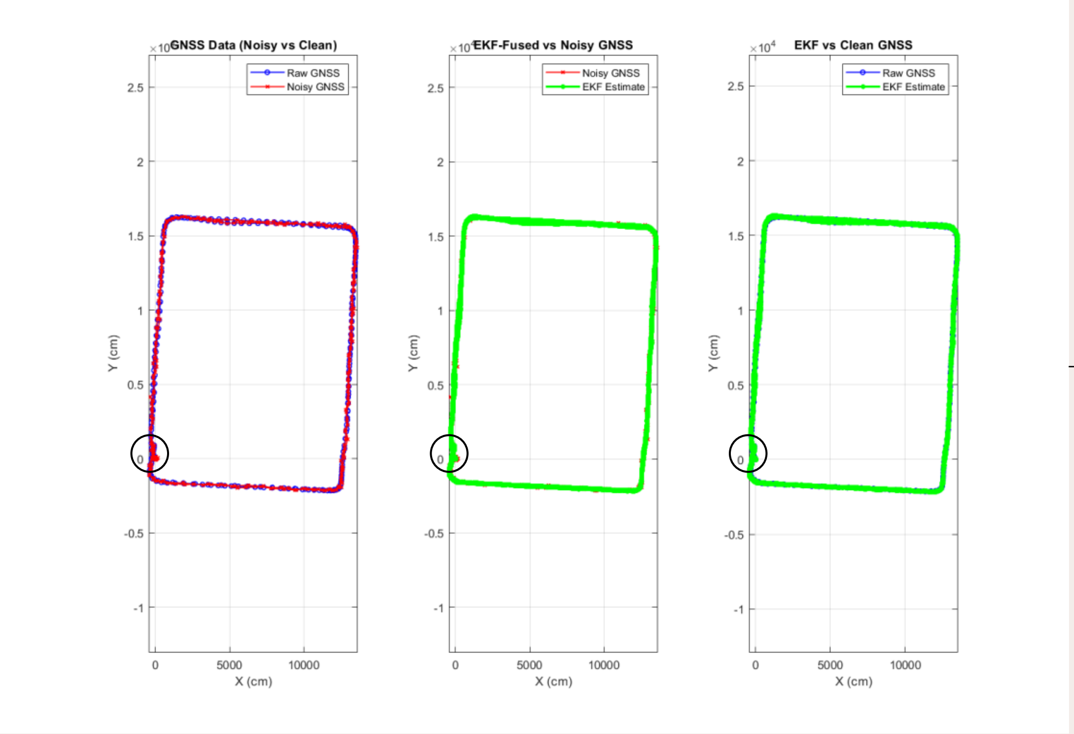
Date of Collection	Total Size	Path length	Sensors
2020/03/14	27.0 GB	1.21 Km	LiDAR/Camera/IMU/SPAN-CPT

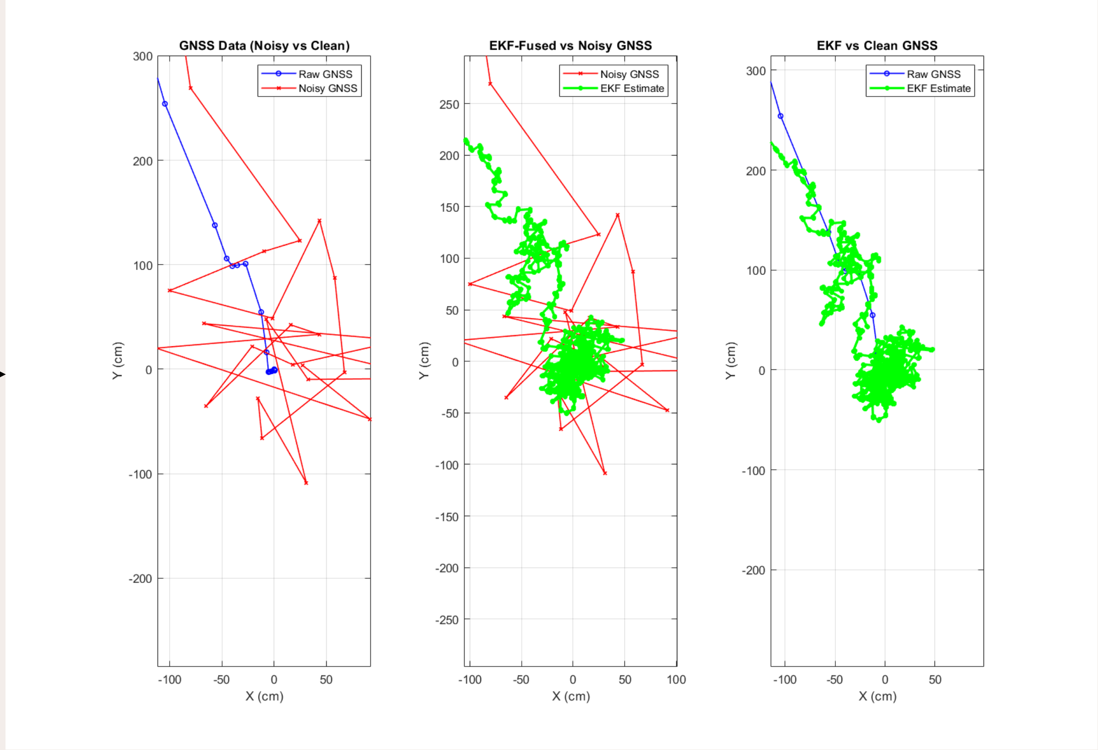
- Download by Dropbox Link:
  - UrbanNav-HK-Data20200314 (ROS)
    - ROSBAG file which includes:
      - 3D LiDAR point clouds: `/velodyne_points`
      - Camera: `/camera/image_color`
      - IMU: `/imu/data`
      - SPAN-CPT: `/novatel_data/inspvax`
  - GNSS (RINEX)
    - GNSS RINEX files, to use it, we suggest to use the RTKLIB



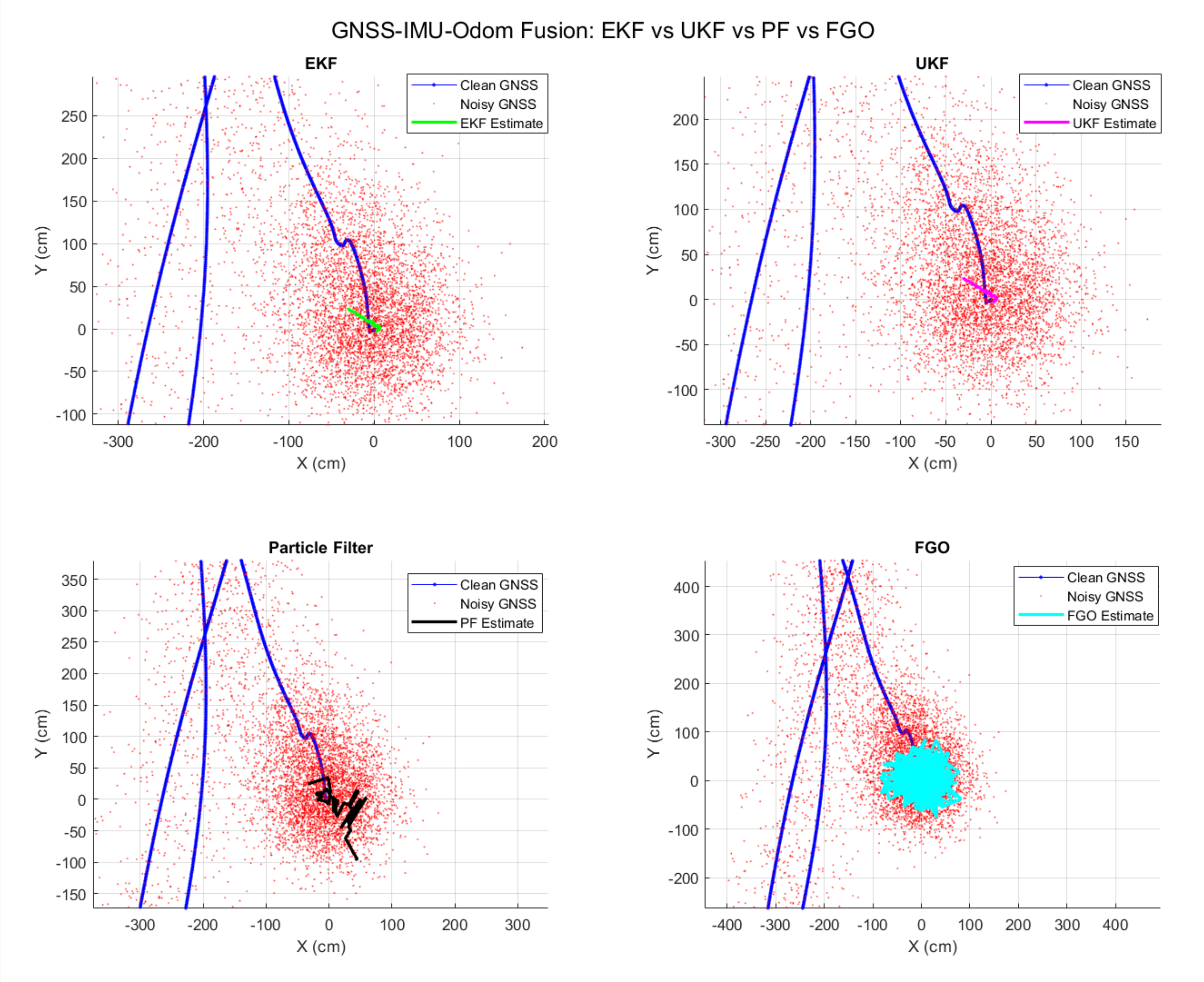
Evaluation & Results

Comparison of Results with noisy, clean GNSS and EKF estimate

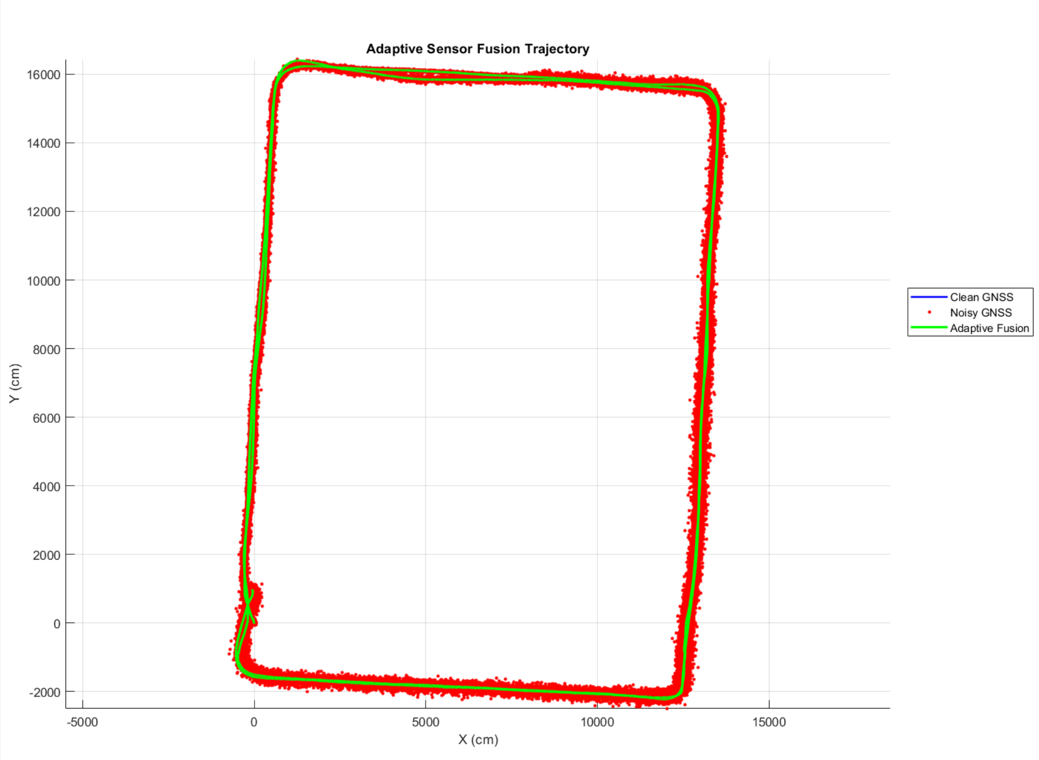


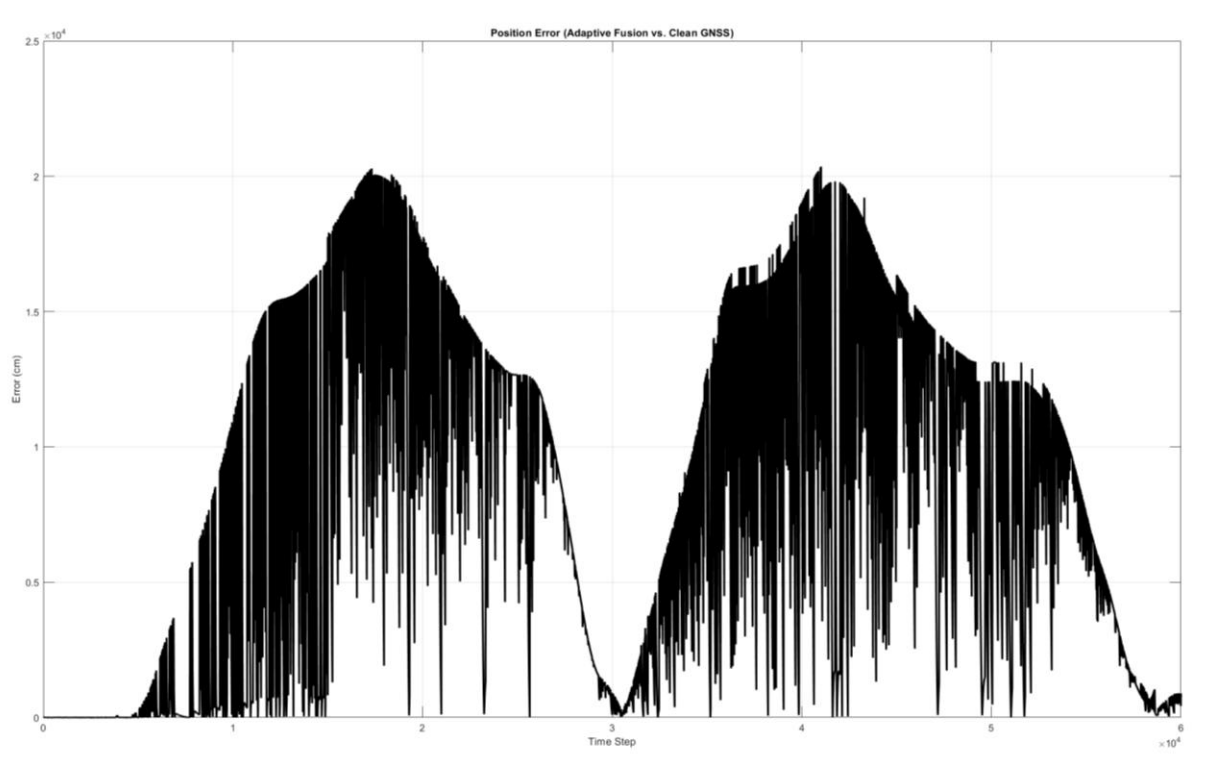


Evaluation & Results

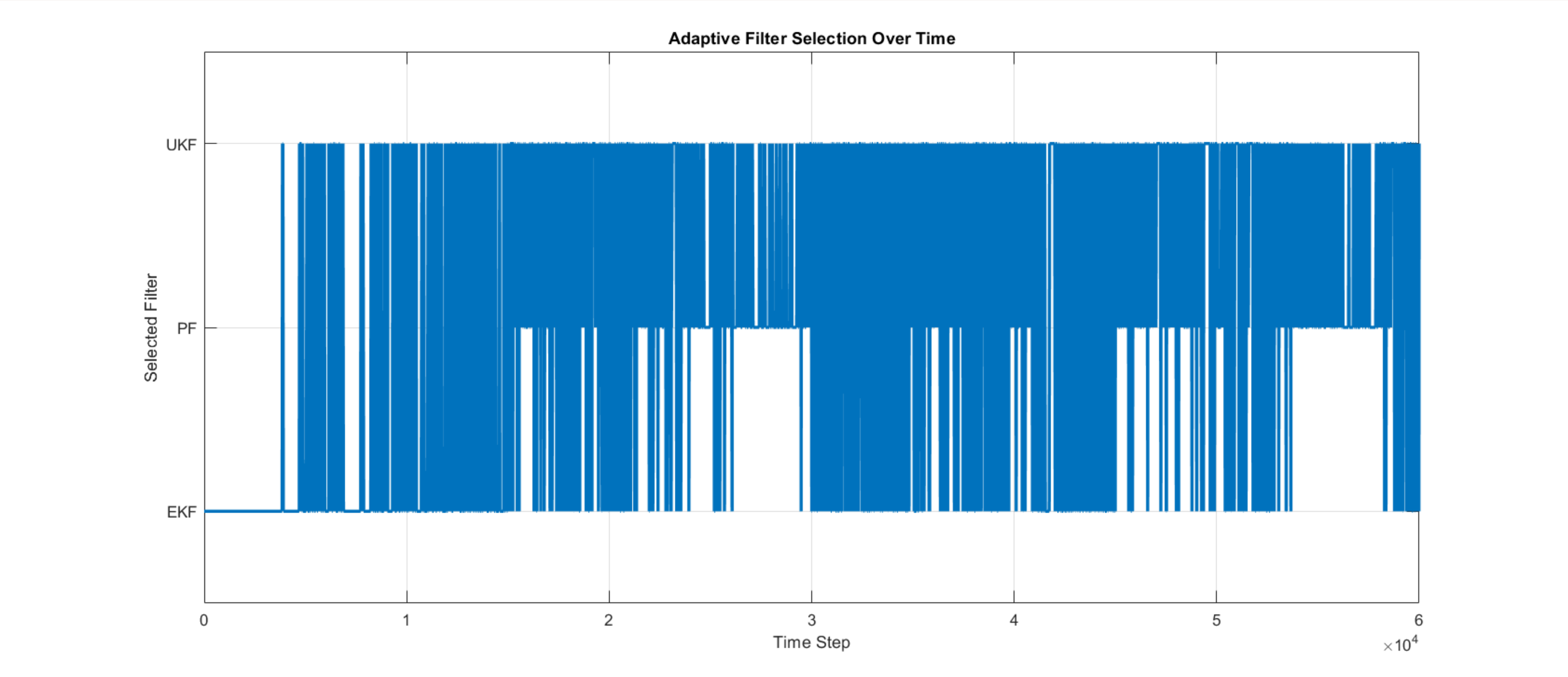


Evaluation & Results





Evaluation & Results



Takeaways

- The localization system was tested using simulated data.
- **Noisy GNSS** data was generated with **time-varying noise levels**.
- The adaptive fusion **reduced localization error** compared to individual sensor estimates.
- **Filter switching** was dynamically adjusted based on sensor reliability.

Key Observations

- **EKF dominated in high-confidence scenarios**.
- **UKF and PF performed better in moderate-to-high noise**.
- **PF was used in low-confidence scenarios**.
- **FGO wasn’t triggered because no extreme noise case were observed.**

Conclusion

This project implemented a robust adaptive sensor fusion system for vehicle localization using GNSS, IMU, and odometer data. The system successfully selected the best filtering approach dynamically to minimize errors and improve positioning accuracy.

What would have been better

Better dataset selection, better noise modelling.

Integration with LiDAR/Camera for better accuracy.

Better research and literature survey is needed

Thank you